

LOGIC and TRUTH TABLES

Logic is the study of how the truth or falsity of a given statement follows (or not!) from the truth of other statements. For example, given the statements

“if G is an Eulerian graph, then no vertex of G has odd degree”

and

“ G is an Eulerian graph”,

we may conclude that

“no vertex of G has odd degree”.

Note. The logic we have used here says nothing about whether the first two statements are actually true or not, but only that **if** they are true, **then** the third is also.

Of course, we could ask whether the first two statements are true (on the basis of other, presumably simpler, statements), and this inquiry would again involve the use of logic.

Definition. A **proposition** is a statement which is unambiguously true or false.

Examples.

- $1 + 1 = 2$;
- $2 + 2 = 3$;
- my birthday is on 29 February;
- there exist infinitely many primes p for which $2^p - 1$ is also prime.

These are **not** propositions:

- $2 + 2$;
- $x^2 + x - 12 = 0$;
- mathematics is more interesting than football;
- do you agree with the previous statement?;
- this sentence is false;
- “yields a false statement when preceded by its own quotation” yields a false statement when preceded by its own quotation.

More complicated expressions can be built up from other propositions by using **logical operators**. The most useful of these are

- “not”, symbolised by $\sim$;
- “and”, symbolised by $\wedge$;
- “or”, symbolised by $\vee$;
- “exclusive or”, symbolised by $\oplus$;
- “if ... then”, symbolised by $\rightarrow$;
- “if and only if”, symbolised by $\leftrightarrow$.

Examples. If p, q, r are propositions then

- $p \wedge (\sim q)$ means “ p and not q ”;
- $(\sim q) \rightarrow (\sim p)$ is the *contrapositive* of $p \rightarrow q$;
- $q \rightarrow p$ is the *converse* of $p \rightarrow q$.

The above propositions, made up of simpler propositions by the use of logical operators, are called *compound propositions*.

If we think of p, q, r as being not specific propositions but variables which may represent any propositions, then we call the combined statements **propositional forms**.

Example. If

p : “I will study tonight” ,

q : “I will go to the pub tonight” ,

r : “I will hand in my assignment on time tomorrow” ,

write the following statements in logical notation.

- (i) I will go to the pub tonight, but I will hand in my assignment on time tomorrow.
- (ii) I will not both study and go to the pub tonight.
- (iii) I will hand in my assignment late tomorrow unless I avoid the pub and study instead tonight.
- (iv) I will only hand in my assignment on time tomorrow if I study tonight.

Answers.

- (i) $q \wedge r$.

Note that “but” is logically the same as “and”.

- (ii) $\sim(p \wedge q)$.

- (iii) $\sim((\sim q) \wedge p) \rightarrow (\sim r)$.

Note that “A unless B” means “A if not B”.

The word “instead” has no logical function and can be ignored.

- (iv) $r \rightarrow p$.

Note that $p \rightarrow r$ may well be true, but it is **not** what the given expression says!

With the same meanings for p, q, r , write the following in ordinary English.

(i) $(\sim p) \wedge (\sim q)$.

(ii) $(p \rightarrow r) \wedge (q \rightarrow \sim r)$.

Answers.

- (i) “I will not study tonight, and I will not go to the pub tonight”.

More concisely, “I will neither study nor go to the pub tonight”.

“Neither ... nor ...” means “not ..., and not ...”.

- (ii) “If I study tonight then I will hand my assignment in on time tomorrow; while if I go to the pub tonight then I will hand my assignment in late”.

Here “while” is just another way of expressing “but” or “and”.

Observe that the truth or falsity of a compound proposition such as $p \vee q$ does not depend on what the propositions p and q *actually* are, but only on whether they are true or false.

The main task of the logic of propositions, therefore, is this: given **truth values** T or F for propositions $p, q, r, \dots$, determine the truth value of a certain compound proposition. This can be done step by step once we know the truth values of the basic combinations $\sim p$, $p \wedge q$, and so on.

Negation. The truth value of $\sim p$ is the opposite of the truth value of p . We can express this by giving the **truth table** for $\sim p$.

p	$\sim p$
T	F
F	T

“**And**”. The conjunction $p \wedge q$ is true when both p and q are true, and false otherwise. Its truth table is

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Problem. Is the above definition circular? If so, how can we fix it?

“Or”. The disjunction $p \vee q$ is true when either p or q , or both, is true, and false otherwise. It has the following truth table.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Note. This is the *inclusive* meaning of “or”.

“Exclusive or”. The exclusive or $p \oplus q$ is true when either p or q is true, but not both, and is false otherwise. Its truth table is as shown.

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

We can use these results to find the truth tables of further compound propositions.

Example. Construct truth tables for $\sim(p \vee q)$ and for $(\sim p) \wedge (\sim q)$.

Solution.

p	q	$p \vee q$	$\sim(p \vee q)$
T	T	T	F
T	F	T	F
F	T	T	F
F	F	F	T

p	q	$\sim p$	$\sim q$	$(\sim p) \wedge (\sim q)$
T	T	F	F	F
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

Definition. Two propositional forms are **logically equivalent** if they have the same truth values for each possible allocation of truth values to the variables in them. We denote the logical equivalence of P and Q by writing $P \Leftrightarrow Q$.

Example. From the above truth tables,

$$\sim(p \vee q) \Leftrightarrow (\sim p) \wedge (\sim q) .$$

This is one of **De Morgan's** laws.

Example. $p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$.

Proof. We find the truth tables as shown.

p	q	r	$q \vee r$	$p \wedge q$	$p \wedge r$	LHS	RHS
T	T	T	T	T	T	T	T
T	T	F	T	T	F	T	T
T	F	T	T	F	T	T	T
T	F	F	F	F	F	F	F
F	T	T	T	F	F	F	F
F	T	F	T	F	F	F	F
F	F	T	T	F	F	F	F
F	F	F	F	F	F	F	F

Since the left hand side and the right hand side have the same truth values in all cases, $p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r)$. This is one of the **distributive** laws.

Two propositional forms can be shown *not* to be equivalent by constructing tables as above and observing that the resulting truth values are *not* always the same, or by producing specific interpretations of the variables in such a way that one of the forms is true and the other false.

Example. Show that $p \wedge q$ and p are not logically equivalent.

Solution. Choose the propositions

$$p : "2 + 3 = 5" \quad \text{and} \quad q : "2 \times 3 = 5" .$$

Then the right hand side is true, while the left hand side is false. Thus $p \wedge q$ and p are not logically equivalent.

Definition. A propositional form that is always true, no matter what the truth values of the propositional variables that occur in it, is called a **tautology**. A propositional form that is always false is called a **contradiction**. A form that is neither a tautology nor a contradiction is called a **contingency**.

Examples.

- $p \vee (\sim p)$ is a tautology.
- $p \wedge (\sim p)$ is a contradiction.
- $p \wedge q$ is a contingency.

To check these assertions, either consider what the propositions mean, or construct their truth tables.

Note.

- Any two tautologies are logically equivalent, since they have the same truth value in all cases. Likewise all contradictions are logically equivalent.
- Conversely, if a proposition is equivalent to a tautology, then the proposition is itself a tautology.

Laws of logical equivalence. Let p, q, r be propositional variables, let $\mathbf{T}$ be a tautology and $\mathbf{F}$ a contradiction. We have already seen the following equivalences:

$$\begin{aligned}\sim(p \vee q) &\Leftrightarrow (\sim p) \wedge (\sim q) \\ p \wedge (q \vee r) &\Leftrightarrow (p \wedge q) \vee (p \wedge r) \\ p \vee (\sim p) &\Leftrightarrow \mathbf{T} \\ p \wedge (\sim p) &\Leftrightarrow \mathbf{F} .\end{aligned}$$

There are many more. *Apart from the different notation they are identical with the laws of set algebra!* Compare, for example,

$$\begin{aligned}\sim(p \vee q) &\Leftrightarrow (\sim p) \wedge (\sim q) \quad \text{and} \quad (A \cup B)^c = A^c \cap B^c \\ p \vee (\sim p) &\Leftrightarrow \mathbf{T} \quad \text{and} \quad A \cup A^c = \mathcal{U} \\ p \wedge (\sim p) &\Leftrightarrow \mathbf{F} \quad \text{and} \quad A \cap A^c = \emptyset .\end{aligned}$$

As with laws of set algebra, logical equivalences come in dual pairs, for example,

$$p \vee (\sim p) \Leftrightarrow \mathbf{T} \quad \text{and} \quad p \wedge (\sim p) \Leftrightarrow \mathbf{F} .$$

Here are some of the most useful logical equivalences.

- Commutative laws:

$$p \wedge q \Leftrightarrow q \wedge p \quad \text{and} \quad p \vee q \Leftrightarrow q \vee p .$$

- Associative laws:

$$(p \wedge q) \wedge r \Leftrightarrow p \wedge (q \wedge r) \\ \text{and} \quad (p \vee q) \vee r \Leftrightarrow p \vee (q \vee r) .$$

- Distributive laws:

$$p \wedge (q \vee r) \Leftrightarrow (p \wedge q) \vee (p \wedge r) \\ \text{and} \quad p \vee (q \wedge r) \Leftrightarrow (p \vee q) \wedge (p \vee r) .$$

- Idempotent laws:

$$p \wedge p \Leftrightarrow p \quad \text{and} \quad p \vee p \Leftrightarrow p .$$

- Double negation law:

$$\sim(\sim p) \Leftrightarrow p .$$

- De Morgan's laws:

$$\sim(p \wedge q) \Leftrightarrow (\sim p) \vee (\sim q) \\ \text{and} \quad \sim(p \vee q) \Leftrightarrow (\sim p) \wedge (\sim q) .$$

- Identity laws:

$$p \wedge \mathbf{T} \Leftrightarrow p \quad \text{and} \quad p \vee \mathbf{F} \Leftrightarrow p .$$

- Domination laws:

$$p \vee \mathbf{T} \Leftrightarrow \mathbf{T} \quad \text{and} \quad p \wedge \mathbf{F} \Leftrightarrow \mathbf{F} .$$

- Laws of negation:

$$p \vee (\sim p) \Leftrightarrow \mathbf{T} \quad \text{and} \quad p \wedge (\sim p) \Leftrightarrow \mathbf{F} .$$

Example. Use the laws of logical equivalence to simplify

$$(p \wedge (\sim q)) \vee (\sim(p \vee q)) .$$

Solution. We have

$$(p \wedge (\sim q)) \vee (\sim(p \vee q))$$

$$\Leftrightarrow (p \wedge (\sim q)) \vee ((\sim p) \wedge (\sim q))$$

(De Morgan's law)

$$\Leftrightarrow (p \vee (\sim p)) \wedge (\sim q) \quad \text{(distributive law)}$$

$$\Leftrightarrow \mathbf{T} \wedge (\sim q) \quad \text{(law of negation)}$$

$$\Leftrightarrow \sim q . \quad \text{(commutative law, and identity law)}$$

“If ... then”. The truth table for the *conditional proposition* or *implication* $p \rightarrow q$ (which we usually read as “if p then q ” or “ p implies q ”) is

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

That is, $p \rightarrow q$ is always true except in the case p true, q false.

Logical equivalences involving $\rightarrow$.

First we show that $p \rightarrow q \Leftrightarrow (\sim p) \vee q$.

p	q	$\sim p$	$p \rightarrow q$	$(\sim p) \vee q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

This is useful since we can now replace $p \rightarrow q$ by $(\sim p) \vee q$ and then apply the 17 logical equivalences that we know.

Example. We can re-express the negation of $p \rightarrow q$ as

$$\begin{aligned}
 \sim(p \rightarrow q) &\Leftrightarrow \sim((\sim p) \vee q) && \text{(above)} \\
 &\Leftrightarrow (\sim(\sim p)) \wedge (\sim q) && \text{(De Morgan's law)} \\
 &\Leftrightarrow p \wedge (\sim q) . && \text{(double negation law)}
 \end{aligned}$$

So to disprove “if p then q ” we have to find a case where p is true and q is false – as we have seen before!!

Example. Also,

$$\begin{aligned}
 (\sim q) \rightarrow (\sim p) &\Leftrightarrow (\sim(\sim q)) \vee (\sim p) && \text{(equivalence for } \rightarrow \text{)} \\
 &\Leftrightarrow q \vee (\sim p) && \text{(double negation law)} \\
 &\Leftrightarrow (\sim p) \vee q && \text{(commutative law)} \\
 &\Leftrightarrow p \rightarrow q .
 \end{aligned}$$

This shows that a statement “if p then q ” is true when its contrapositive is true, and false when its contrapositive is false.

Example. Are $p \rightarrow q$ and $q \rightarrow p$ logically equivalent?

Answer. No. We construct the truth table.

p	q	$p \rightarrow q$	$q \rightarrow p$
T	T	T	T
T	F	F	T
F	T		
F	F		

We might as well stop here: the truth values will not always be the same, so $p \rightarrow q$ is not logically equivalent to $q \rightarrow p$. This shows that the truth values of a proposition and its converse are not always the same.

Note that the *non-equivalence* of two propositional forms would normally be very difficult to prove by “algebraic” methods.

Example. An example involving $\rightarrow$ and tautologies:

$$(p \wedge q) \rightarrow p$$

is a tautology.

Proof by truth table:

p	q	$p \wedge q$	$(p \wedge q) \rightarrow p$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

Exercise. Prove this by using logical equivalences to show

$$(p \wedge q) \rightarrow p \Leftrightarrow \dots \Leftrightarrow \mathbf{T} .$$

“If and only if”. The *biconditional* proposition $p \leftrightarrow q$ is true when p and q are both true or both false, and is false otherwise. Its truth table is

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

We can check by a truth table that

$$p \leftrightarrow q \Leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p) ,$$

which corresponds with our previous “informal” definition of “if and only if”.

Note carefully the difference between the symbols $\Leftrightarrow$ and $\leftrightarrow$. If P and Q are two propositional forms, then

$$P \Leftrightarrow Q$$

is a statement which tells you something about these two forms, whereas

$$P \leftrightarrow Q$$

is a combination of the two propositional forms into a single form.

There is, however, a close relationship between logical equivalence and the biconditional proposition.

Theorem. Two propositional forms P and Q are logically equivalent if and only if $P \leftrightarrow Q$ is a tautology.

Proof. Suppose that $P \leftrightarrow Q$ is a tautology. Then for all truth values of the constituent variables, the truth value of $P \leftrightarrow Q$ is T . Thus for all truth values of the variables, P and Q have the same truth value. So by definition, $P \Leftrightarrow Q$.

Conversely, if $P \Leftrightarrow Q$ then in all cases P and Q have the same truth values; so in all cases $P \leftrightarrow Q$ is true. That is, $P \leftrightarrow Q$ is a tautology.

Example. We know that

$$\sim(p \vee q) \Leftrightarrow (\sim p) \wedge (\sim q) ;$$

therefore

$$(\sim(p \vee q)) \leftrightarrow ((\sim p) \wedge (\sim q))$$

is a tautology. Check this by algebra or by a truth table.

Definition. Let P and Q be propositional forms. Suppose that in any case where P is true, Q is also true. Then we say that P **logically implies** Q and write $P \Rightarrow Q$.

Example. Consider the propositional forms P and Q , defined to be

$$(p \rightarrow q) \wedge r \quad \text{and} \quad p \rightarrow (q \wedge r) .$$

respectively. Does P logically imply Q ? Does Q logically imply P ?

Solution. We draw up a truth table for both propositional forms.

p	q	r	$p \rightarrow q$	$q \wedge r$	$(p \rightarrow q) \wedge r$	$p \rightarrow (q \wedge r)$
T	T	T	T	T	T	T
T	T	F	T	F	F	F
T	F	T	F	F	F	F
T	F	F	F	F	F	F
F	T	T	T	T	T	T
F	T	F	T	F	F	T
F	F	T	T	F	T	T
F	F	F	T	F	F	T

The cases where P is true occur in rows 1, 5 and 7; in these cases Q is always true. That is, if P is true then Q is also true. Therefore P logically implies Q .

On the other hand, if Q is true then P is not always true, for example in row 8. Therefore Q does not logically imply P .

As with logical equivalence (compare page 101), be sure you understand the difference between $\Rightarrow$ and $\rightarrow$. If P and Q are propositional forms then

$$P \Rightarrow Q$$

is a statement which tells you something about these two propositions (namely, that if P is true then Q must also be true), while

$$P \rightarrow Q$$

is a single propositional form built up from these two. Once again, however, though the two symbols are not identical, there is a close relationship between them.

Theorem. Let P and Q be propositional forms. Then P logically implies Q if and only if $P \rightarrow Q$ is a tautology.

Proof. Suppose that $P \rightarrow Q$ is a tautology. Then it is never false; so there is no case where P is true and Q is false. That is, in every case where P is true, Q is also true. But this is exactly what is meant by $P \Rightarrow Q$.

Conversely, suppose that $P \Rightarrow Q$ and consider the truth values of $P \rightarrow Q$.

Case 1, P is true. Since $P \Rightarrow Q$ we see that Q is also true. Therefore $P \rightarrow Q$ is true.

Case 2, P is false. Then $P \rightarrow Q$ is true, regardless of the truth value of Q .

Thus $P \rightarrow Q$ is true in every case, that is, it is a tautology.

Example. Show that

$$((p \rightarrow q) \wedge p) \Rightarrow q .$$

Solution. We draw up a truth table for $\text{LHS} \rightarrow \text{RHS}$.

p	q	$p \rightarrow q$	$(p \rightarrow q) \wedge p$	$((p \rightarrow q) \wedge p) \rightarrow q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

Since the final result is always T we see that

$$((p \rightarrow q) \wedge p) \rightarrow q$$

is a tautology. That is,

$$((p \rightarrow q) \wedge p) \Rightarrow q .$$

Exercise. Repeat this problem using the method of page 103. (The working is almost identical.)

Theorem. Let P and Q be propositional forms. Then $P \Leftrightarrow Q$ if and only if both $P \Rightarrow Q$ and $Q \Rightarrow P$.

Proof. Exercise.

VALID AND INVALID ARGUMENTS

Recall the argument we saw before:

“If G is an Eulerian graph, then no vertex of G has odd degree.

G is an Eulerian graph.

Therefore, no vertex of G has odd degree.”

The deduction made here uses the **rule of inference**

$$\frac{p \rightarrow q \quad p}{\therefore q}$$

Such an argument is **valid** because the **conclusion** (the last statement) *must always be true*, provided that the **hypotheses** (earlier statements) are true.

This particular rule of inference is commonly known as *modus ponens*.

The following supposed rule of inference is **invalid**,

$$\frac{p \rightarrow q \quad q}{\therefore p},$$

since in some cases the hypotheses will be true and the conclusion false. For example, take

$$\begin{aligned} p &= \text{“I live in Queensland”} \\ q &= \text{“I live in Australia”} . \end{aligned}$$

In fact the above (invalid) argument is simply the converse fallacy.

Note carefully that the conclusion of a valid argument is *not always true*! For example,

“If my pet Fluffy drinks milk, then she is a dinosaur.
My pet Fluffy drinks milk.
Therefore, my pet Fluffy is a dinosaur.”

To be sure of reaching a true conclusion we need a valid argument **and** true hypotheses.

Another rule of inference, known as *modus tollens*, is

$$\frac{\begin{array}{c} p \rightarrow q \\ \sim q \end{array}}{\therefore \sim p .}$$

For example,

“If you kick this goal, then we will win the match.

We did not win the match.

Therefore, you did not kick the goal.”

Essentially, this is proof by use of the contrapositive.

The *hypothetical syllogism* is the rule of inference

$$\frac{p \rightarrow q \quad q \rightarrow r}{\therefore p \rightarrow r}.$$

For example,

“If I study hard, then I will earn a degree.

If I earn a degree, then I will get a good job.

Therefore, if I study hard, then I will get a good job.”

We can use truth tables to check that the rule of inference *modus ponens*

$$\frac{p \rightarrow q \quad p}{\therefore q}$$

is a valid method of argument. First we construct a truth table showing truth values of the hypotheses and the conclusion.

p	q	$p \rightarrow q$	p	q
T	T	T	T	T
T	F	F	T	F
F	T	T	F	T
F	F	T	F	F

The only case in which both the hypotheses are true occurs in the first row; and here the conclusion is also true. Hence, the argument is valid.

To confirm that the converse fallacy

$$\frac{p \rightarrow q \quad q}{\therefore p}$$

does *not* constitute a valid argument, we again construct a truth table.

p	q	$p \rightarrow q$	q	p
T	T	T	T	T
T	F	F	F	T
F	T	T	T	F
F	F	T	F	F

Here the hypotheses are true in the first and third rows; but in these rows the conclusion is *not always true*. In particular, the third row exhibits a case where the hypotheses may be true and the conclusion false. Thus, this is not a valid argument.

Valid arguments and tautologies. Observe that the truth table we used on page 110 to confirm that *modus ponens* is valid is very similar to that used on page 105, showing that $(p \rightarrow q) \wedge p$ logically implies q .

Theorem. An argument

$$\frac{\begin{array}{c} P_1 \\ P_2 \\ \vdots \\ P_n \end{array}}{\therefore Q}$$

is valid if and only if $(P_1 \wedge P_2 \wedge \cdots \wedge P_n) \Rightarrow Q$, that is, if and only if the proposition

$$(P_1 \wedge P_2 \wedge \cdots \wedge P_n) \rightarrow Q \quad \otimes$$

is a tautology.

Proof. Suppose the given argument is valid. Now the proposition $P_1 \wedge P_2 \wedge \cdots \wedge P_n$ is either true or false.

Case 1, $P_1 \wedge P_2 \wedge \cdots \wedge P_n$ is false. Then $\otimes$ is true (regardless of the truth value of Q).

Case 2, $P_1 \wedge P_2 \wedge \cdots \wedge P_n$ is true. Then Q is true, so $\otimes$ has the form $a \rightarrow b$, with a, b both true; and this is a true statement.

Thus $\otimes$ has the truth value T in all cases, and is therefore a tautology.

Conversely, suppose that the argument is invalid. Then there exists a case in which all of $P_1, P_2, \dots, P_n$ are true yet Q is false. In this case $\otimes$ is of the form $a \rightarrow b$ with a true, b false; so $\otimes$ is false. Since $\otimes$ is not always true, it is not a tautology.

Things to learn from this proof.

- Structure of an “if and only if” proof.
- Proof by division into cases.
- Proof by contrapositive.

Another valid argument form is

$$\frac{\begin{array}{l} p \vee q \\ p \rightarrow r \\ q \rightarrow r \end{array}}{\therefore r .}$$

We prove this by showing that

$$((p \vee q) \wedge (p \rightarrow r) \wedge (q \rightarrow r)) \rightarrow r$$

is a tautology. For brevity we write it as $P \rightarrow r$.

p	q	r	$p \vee q$	$p \rightarrow r$	$q \rightarrow r$	P	$P \rightarrow r$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	T	T	T	T	T
T	F	F	T	F	T	F	T
F	T	T	T	T	T	T	T
F	T	F	T	T	F	F	T
F	F	T	F	T	T	F	T
F	F	F	F	T	T	F	T

Thus $P \rightarrow r$ is a tautology, and the argument form is valid.

In fact this argument form is simply proof by **division into cases**!

Theorem. The rule of inference

$$\frac{(\sim p) \rightarrow \mathbf{F}}{\therefore p}$$

is valid.

Proof. By the laws of logical equivalence,

$$\begin{aligned} ((\sim p) \rightarrow \mathbf{F}) \rightarrow p &\Leftrightarrow (\sim(\sim p) \vee \mathbf{F}) \rightarrow p && \text{(equivalence for } \rightarrow \text{)} \\ &\Leftrightarrow (p \vee \mathbf{F}) \rightarrow p && \text{(double negation)} \\ &\Leftrightarrow p \rightarrow p && \text{(identity law)} \\ &\Leftrightarrow (\sim p) \vee p && \text{(equivalence for } \rightarrow \text{)} \\ &\Leftrightarrow \mathbf{T} . && \text{(law of negation)} \end{aligned}$$

This is a formal verification of the method of **proof by contradiction**.

Problem. Suppose it is known that

“If I don’t leave home early I will miss the bus.
 If I run fast I will catch the bus.
 I caught the bus.”

Did I leave home early?

Solution. Let’s see if we can prove that I left home early. First we write the argument in symbolic form. Define the propositions

h = “I left home early”
 c = “I caught the bus”
 r = “I ran fast”.

We need to decide whether or not the argument form

$$\frac{\begin{array}{l} (\sim h) \rightarrow (\sim c) \\ r \rightarrow c \\ c \end{array}}{\therefore h}$$

is valid.

Method 1, rules of inference. The first statement is logically equivalent to its contrapositive $c \rightarrow h$. But we also know from the third statement that c is true. Therefore by *modus ponens* h is also true. Hence the argument is valid and **yes**, I did leave home early.

... continued

Method 2, truth table. We show that

$$((\sim h) \rightarrow (\sim c)) \wedge (r \rightarrow c) \wedge c$$

logically implies h . Denote the above statement by H .

h	c	r	$(\sim h) \rightarrow (\sim c)$	$r \rightarrow c$	c	H	h
T	T	T	T	T	T	T	T
T	T	F	T	T	T	T	T
T	F	T	T	F	F	F	T
T	F	F	T	T	F	F	T
F	T	T	F	T	T	F	F
F	T	F	F	T	T	F	F
F	F	T	T	F	F	F	F
F	F	F	T	T	F	F	F

Now H is true in the first two rows; and in these cases h is true too. Therefore

$$(((\sim h) \rightarrow (\sim c)) \wedge (r \rightarrow c) \wedge c) \Rightarrow h ;$$

this proves that the argument is valid and I must have left home early.

Method 3, logical equivalences. Use the laws of logical equivalence to prove that

$$(((\sim h) \rightarrow (\sim c)) \wedge (r \rightarrow c) \wedge c) \rightarrow h$$

is a tautology.

Problem. Suppose again that

“If I don’t leave home early I will miss the bus.

If I run fast I will catch the bus.

I caught the bus.”

Did I run fast?

Solution. Defining the propositions h, c and r as above, we now have to consider the argument form

$$\begin{array}{c} (\sim h) \rightarrow (\sim c) \\ r \rightarrow c \\ c \\ \hline \therefore r . \end{array}$$

If we try to deduce r from the given statements we don’t seem to have any success. This leads us to suspect that the argument form is **invalid**. We can check this by using a truth table to show that there is a case where the three hypotheses are true but the conclusion is false. Since we only need one case there is no need to work out the whole truth table; a bit of trial and error shows that if we take

$$h \text{ true, } c \text{ true and } r \text{ false}$$

then the assumptions are true but r is false. Thus the argument is invalid and we **cannot** conclude that I ran fast.

... continued

Does this mean that I didn't run fast? Consider the argument form

$$\begin{array}{c}
 (\sim h) \rightarrow (\sim c) \\
 r \rightarrow c \\
 c \\
 \hline
 \therefore \sim r .
 \end{array}$$

In the same way (*exercise!*) we can show that this argument is not valid either. So, given the three facts we started with, did I run fast? The answer is that **we cannot tell** because insufficient information is given.

Exercise. Discuss the validity of the following argument.

“Any argument which obeys all the laws of logic is valid.

This argument does not obey all the laws of logic.

Therefore, this argument is not valid.”